

INCIDENT CASE FILE & VESSEL ATTRIBUTION AUDIT

LEGAL DETERRENCE & MARPOL ANNEX I EVIDENCE DOSSIER

Case Identifier:	CASE-2026-MUMBAI-001	Generated Timestamp:	2026-09-11T20:00:16.512640+00:00
Target SAR Scene:	S1A_IW_GRDH_1SDV_20260515T060000_MUMBAI	H3 Grid Resolution:	Res 7 (~5.16 km² / cell)
Attribution Method:	Lagrangian Backtracking + H3 Hash Join	Null Result Status:	CONFIRMED (Culprit Identified)

[CHAIN OF CUSTODY] SHA-256 Tamper-Evident Input Dataset Hash:

d9845cb370ed3639f7e4833f6270af84cf8b37e2684a18385872010f1f322850

This digital fingerprint verifies identical sensor inputs and model parameters were used for reproducible judicial attribution.

STAGE 4 (PART B) // DARK VESSEL DETECTION (RADAR vs AIS)

WARNING: COVERT / NON-BROADCASTING VESSEL DETECTED NEAR ORIGIN

CFAR Detection ID: CFAR_DARK_002 | Position: 19.2800 N, 71.9000 E | Timestamp: 2026-05-15T06:00:00Z

AIS Broadcast: **FALSE (TRANSPONDER INACTIVE)** | Proximity Corridor Cell: 8742da462fffff

Forensic Interpretation: Physical vessel confirmed via Sentinel-1 SAR backscatter. Absence of AIS transponder broadcast within spill head indicates deliberate evasive operation or dark illicit discharge.

STAGE 4 (PART A) // MULTI-FACTOR VESSEL ATTRIBUTION RANKING

Rank	Vessel ID / Name	Corridor (40%)	Heading (25%)	Speed (20%)	AIS Gap (15%)	Total Score	Attribution Decision
#1	IND_TANKER_412 MMSI_419000101	0.14	1.00	1.00	1.00	0.657	EXONERATED

STAGE 2 // PHYSICAL BACKTRACK & ENVIRONMENTAL PARAMETERS

Hydrodynamic Model:	HYCOM-GLBv0.08 (OPeNDAP)	Atmospheric Forcing:	ERA5-Reanalysis (Copernicus CDS)
Wind Drift Factor:	3.0% of 10m Wind	Lagrangian Particles:	1000 ensemble tracers
Look-Alike Wind Gate:	2.0 - 12.0 m/s (Passed)	Diffusion Coeff:	1.0 m²/s stochastic

Timestep	Timestamp (UTC)	H3 Cells Count	Peak Density Cell	Spatial Role
t0 (Observation)	2026-05-15T06:00:00Z	6	8742da54efffff (227 p)	Slick Observation Footprint
t - 6h	2026-05-15T00:00:00Z	11	8742da541fffff (171 p)	Advection Corridor (Mid)
t - 12h	2026-05-14T18:00:00Z	14	8742da563fffff (141 p)	Estimated Discharge Window
t - 24h	2026-05-14T06:00:00Z	18	8742dacc2fffff (105 p)	Corridor Bound / Dispersion Limit